Solutions to Exercises from Chapter 4
Reinforcement Learning (2nd Edition) by Sutton & Barto

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Exercise 4.1

Under the equiprobable random policy, the action-value function $q_\pi(s, a)$ is computed as:

$$q_\pi(s, a) = \sum_{s',r} p(s', r \mid s, a)[r + \gamma v_\pi(s')]$$

Using Example 4.1 and the associated gridworld:

- $q_\pi(11, \text{down}) = -1 + \gamma v_\pi(11) = -1 + 0.9 \times 0 = -1$
- $q_\pi(7, \text{down}) = -1 + \gamma v_\pi(11) = -1 + 0.9 \times 0 = -1$

Exercise 4.2

Case 1: New state 15 is added, but no other changes:

$$v_\pi(15) = \frac{1}{4}[v_\pi(12)+v_\pi(13)+v_\pi(14)+v_\pi(15)] \Rightarrow v_\pi(15) = \frac{1}{3}[v_\pi(12)+v_\pi(13)+v_\pi(14)] \Rightarrow v_\pi(15) = \frac{1}{3}(0+0+0) = 0$$

Case 2: Action "down" from state 13 leads to 15:

$$v_\pi(13) = \frac{1}{4}[v_\pi(9) + v_\pi(12) + v_\pi(14) + v_\pi(15)]$$

Now 13's value depends on 15 and 15 depends on 13. Solving this recursive system yields $v_\pi(15) > 0$.

Exercise 4.3

$$\begin{aligned}
v_\pi(s) &= \sum_a \pi(a|s) q_\pi(s, a) \\
q_\pi(s, a) &= \sum_{s',r} p(s', r|s, a)[r + \gamma v_\pi(s')] \\
q_\pi(s, a) &= \sum_{s',r} p(s', r|s, a)[r + \gamma \sum_{a'} \pi(a'|s') q_\pi(s', a')]
\end{aligned}$$

Exercise 4.4

Modified Step 3 (Policy Improvement):

- Let $\pi'(s) \in \arg\max_a q_\pi(s, a)$
- If $\pi' = \pi$, then stop.
- Else, update $\pi \leftarrow \pi'$.

Exercise 4.6

For ϵ-soft policies:

- **Step 3 (Policy Improvement):**

$$\pi(a|s) = \begin{cases} 1 - \epsilon + \frac{\epsilon}{|A(s)|}, & \text{if } a = \arg\max_a q(s,a) \\ \frac{\epsilon}{|A(s)|}, & \text{otherwise} \end{cases}$$

- **Step 2 (Policy Evaluation):** Evaluate using:

$$v(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)[r + \gamma v(s')]$$

- **Step 1 (Initialization):** Use uniform ϵ-soft policy.

Exercise 4.8

Recall the iterative policy evaluation equation (Equation 4.1):

$$v_{k+1}(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r \mid s,a) \left[r + \gamma v_k(s')\right]$$

For a deterministic policy π, the agent always chooses the same action in each state. That is, for each state s, there is a single action $a = \pi(s)$ such that:

$$\pi(a|s) = \begin{cases} 1 & \text{if } a = \pi(s) \\ 0 & \text{otherwise} \end{cases}$$

Thus, the evaluation equation simplifies to:

$$v_{k+1}(s) = \sum_{s',r} p(s',r \mid s,\pi(s)) \left[r + \gamma v_k(s')\right]$$

This is a deterministic update based on the fixed policy π. Now, addressing the questions:

- **Will the values change?**
  Yes, the values will change during iterations as the evaluation process proceeds. Starting from an initial guess $v_0(s)$, the updates will continue to adjust $v_k(s)$ until convergence is reached. At convergence, $v_k(s) = v_\pi(s)$ for all s.
- **What does this say about the relationship between the value function and the policy?**
  This indicates that the value function is a consequence of the policy. For a given deterministic policy π, there exists a unique value function v_π representing the expected return from each state while following π. Therefore, the policy fully determines the value function.

Conclusion: The values will change iteratively

Exercise 4.10

$$q_{k+1}(s,a) = \sum_{s',r} p(s',r|s,a) \left[r + \gamma \max_{a'} q_k(s',a')\right]$$

Gambler's Problem: Curious Optimal Policy Explanation

In the Gambler's Problem, the agent's goal is to reach \$100. The optimal policy is:

- At $s = 50$: betting all (\$50) gives a direct chance to reach \$100 with:

$$v(50) = p_h$$

- At $s = 51$: max bet is \$49, leading to:
  - Win: \$100
  - Lose: \$2 (low value)

  Safer smaller bets keep the agent in high-value zones and avoid risky positions.

Conclusion: Though the policy appears inconsistent, it's globally optimal. At 50, betting all is best. At 51, risk-averse moves improve long-term success.